Product and Quotient Rules Practice

Find the derivative of each function using the appropriate rules.

1. $f(x)=\left(3x-2x^2\right)\left(5+4x\right)$

2. $f(x)=\left(\frac{1}{x}+1\right)\left(x-1\right)$

3. $f(x)=\left(2t^2-3\right)\left(4-t^2-t^4\right)$

4. $f(x)=\left(p^3-2\right)^2$

5. $f(x)=\frac{x-1}{2x+3}$

6. $f(x)=\frac{2x^2-4x+3}{2-3x}$

7. $f(x)=\frac{7}{x^3}$

8. $f(x)=\frac{4x^2-3x}{8\sqrt{x}}$

9. $f(x)=\left(\frac{x-3}{x+4}\right)\left(x^2+2x+1\right)$

10. $f(x)=\frac{\left(1-2x\right)\left(3x+2\right)}{5x-4}$